

Assignment: Day 6

Introduction to Network Devices

Q1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

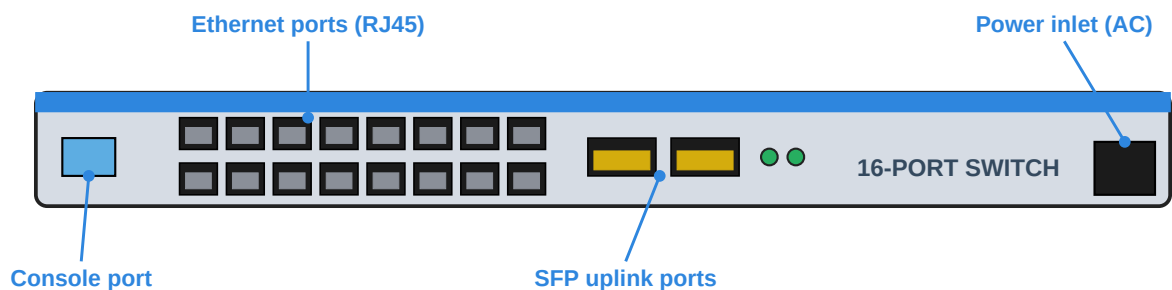
Ans:

Device	Main functions	Typical ports	OSI layer
Hub	Repeats every incoming signal to all other ports. It has no intelligence, so all devices share the same bandwidth and collisions can happen. Rarely used today.	4 to 8 Ethernet (RJ45) ports, power jack	Layer 1 (Physical)
Switch	Learns MAC addresses and sends data only to the port of the intended device. It gives every wired device (PC, printer, smart TV) its own bandwidth.	8 to 24 Ethernet ports (10/100/1000 Mbps), SFP uplink (some), console port (managed models), power	Layer 2 (Data Link)
Router	Connects the home network to the internet and routes packets using IP addresses. Also provides DHCP (gives IPs), NAT and usually a basic firewall and built-in WiFi.	1 WAN/Internet port, 4 LAN Ethernet ports, USB, power jack, reset button, WiFi antennas	Layer 3 (Network)
Firewall	Checks incoming and outgoing traffic against security rules and blocks unwanted or harmful traffic. In homes it is usually built into the router.	WAN port, LAN ports, DMZ port, management and console port, USB, power	Layers 3 to 7
Access Point (AP)	Turns the wired network into WiFi so phones and laptops can connect wirelessly. It extends WiFi coverage to more rooms.	1 Ethernet/PoE port, power (DC) jack, reset button, console port (some), antennas (internal or external)	Layer 2 (Data Link)

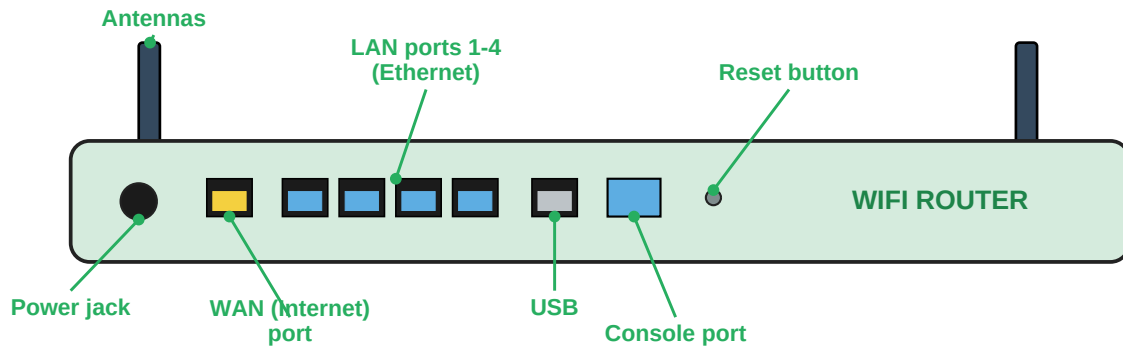
Q2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Ans: Four devices are labelled below with their ports.

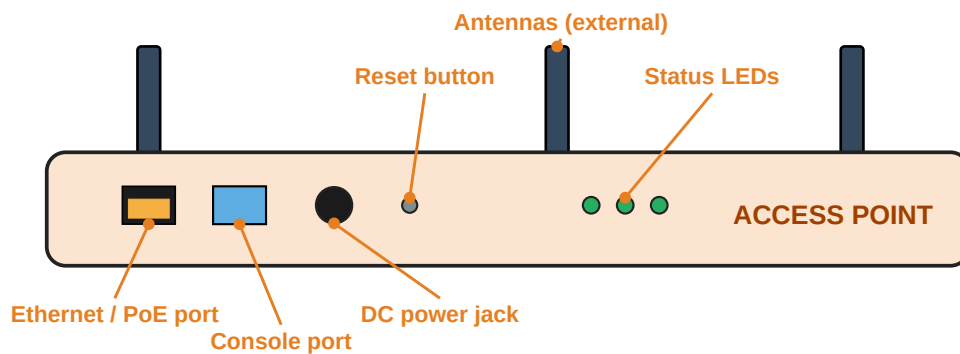
1. Switch



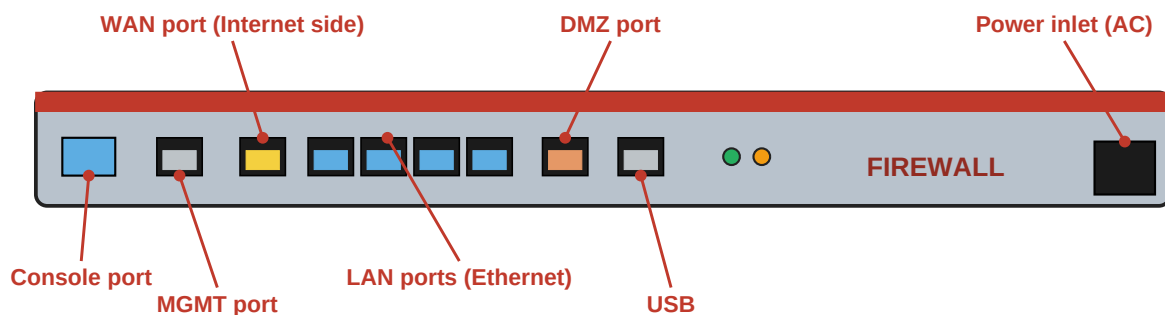
2. WiFi Router



3. Access Point



4. Firewall



Q3. Pick a popular app you use daily and describe in 3-4 lines how at least two different network devices play a role when you use that app.

Ans: App: **Instagram**. When I open Instagram and scroll reels on my phone, the phone first connects wirelessly to the **access point** (the WiFi), which receives my request for new videos and passes it to the wired network. The **router** then sends this request out to the internet, finds Instagram's server using IP addresses, and brings the video data back to my phone. The **firewall** built into the router checks this traffic and blocks anything unsafe.

Q4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Ans: Possible reason: The access points are overloaded. Every access point can only serve a limited number of devices at the same time. In a college, hundreds of students connect phones and laptops to the same few older access points, especially in classrooms and the library. The access point then has to share its limited airtime between too many users, so the speed drops, and when its client limit is reached or the signal is weak in far corners, devices get disconnected.

How upgrading one device helps: Replacing the old access points with newer WiFi 6 dual-band or tri-band access points, and placing more of them (for example one per classroom), would solve this. WiFi 6 uses technologies like OFDMA and MU-MIMO to serve many devices at once, and having more access points reduces the number of users per access point and improves coverage, so the WiFi becomes faster and more stable.